

AI for Clinical Study Reports and Submission

Yilong Zhang

Nan Xiao

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Welcome

Welcome to AI for Clinical Study Workflows.

The goal of this book is to demonstrate the SDLC of building and maintain AI-first workflow for clinical trial development.

The book focus on showing general principals instead of detail implementations as that highly depends on the organization's AI strategy.

Whether you are new to AI-native workflows or already working in clinical reporting, you will find small, reusable examples you can adopt right away, along with patterns and checklists for organizing work for regulatory submission and effective team collaboration.

This is a work-in-progress draft.

Preface

In this book

This book is designed for people who are interested in using AI for clinical development. Each part of the book makes certain assumptions about the readers' background:

- Part 1, TBD
- Part 2, TBD
- Part 3, TBD
- Part 4, TBD

Authors and contributors

This document is a collaborative effort maintained by a community. As you read through it, you also have the opportunity to contribute and enhance its quality. Your input and involvement play a vital role in shaping the excellence of this document.

- Authors: made significant contributions to at least one chapter, constituting the majority of the content.

[Yilong Zhang](#), [Nan Xiao](#),

Part I

Principles

1 Principles overview

This chapter defines the principles for designing an AI-first clinical study workflow.

The goal is not to replace clinical statisticians or programmers. The goal is to design work so an AI agent can perform a structured first pass, inspect real artifacts with tools, prepare evidence-backed findings, and ask humans to make accountable decisions.

1.1 What is an AI-first workflow?

An AI-first workflow represents work as inspectable artifacts, tracks explicit state, triggers agent action through events and rules, grounds reasoning in tools and evidence, and keeps humans accountable for judgment.

In this book, the agent is an **independent reviewer assistant**.

It can inspect, compare, prepare findings, rerun checks, and escalate judgment calls. It does not make final statistical decisions, sign off deliverables, or replace the responsible statistician or programmer.

1.2 Root principle: Trustworthy assistance

The first principle of an AI-first clinical workflow is **trustworthy assistance**.

Trustworthy assistance means the agent's work can be inspected, repeated, challenged, and reviewed. The agent should not rely on hidden memory, unsupported claims, or vague reasoning. It should work from visible artifacts, preserve explicit state, act through defined events and rules, use tools to gather evidence, and escalate judgment to accountable humans.

These principles do not replace organizational validation, quality systems, access control, audit trail requirements, or regulatory accountability. They describe how to design the working pattern so AI assistance remains inspectable and reviewable.

The remaining principles describe what trustworthy assistance requires in practice.

A workflow is trustworthy when its inputs are visible, its actions are traceable, its findings are evidence-backed, and its limits are clear.

1.3 Principle 1: Inspectable artifacts

Trust requires the agent and humans to inspect the same source materials.

The workflow should use artifacts that humans and AI agents can inspect. Artifacts should be readable, structured where possible, versioned, citable, and diffable.

Examples:

- SAP sections should be searchable and citable.
- ADaM metadata should expose dataset names, variable names, labels, types, codelists, and analysis flags.
- TLF outputs should have stable table IDs, titles, footnotes, and source references.
- Findings should be structured records, not only narrative comments.

If the agent cannot inspect the work, it cannot reliably help with the work.

1.4 Principle 2: Explicit state

Trust requires knowing what changed, what was checked, what was decided, and what remains unresolved.

The workflow should remember through artifacts, not through chat.

State should show:

- which artifact versions are current;
- what changed;
- what was checked;
- what findings were produced;
- what humans decided and why;
- what remains unresolved.

This matters because clinical study work changes repeatedly. Without explicit state, every agent run becomes a disconnected one-time review. With explicit state, the workflow can report which findings are new, repeated, fixed, acknowledged, or still open.

The workflow state should be visible, reusable, and reviewable.

1.5 Principle 3: Events and rules

Trust requires predictable triggers, not random or purely chat-driven action.

The workflow should not depend only on manual prompting.

An **event** says something changed. A **rule** decides what the agent should do next.

Event	Rule
SAP section updated	rerun affected SAP-to-TLF checks
ADaM metadata changed	rerun checks using changed variables
ADaM dataset refreshed	rerun affected population and value checks
TLF output regenerated	rerun affected TLF checks
Human asks for recheck	run the requested review
Another AI agent changes a deliverable	verify downstream impact

Without explicit triggers, review coverage depends on who remembered to ask. Events and rules make the workflow easier to repeat, explain, and audit.

Events trigger attention; rules trigger action.

1.6 Principle 4: Tools and evidence

Trust requires findings grounded in real SAP, ADaM, and TLF artifacts, not model memory or guesses.

The LLM should not guess what is in the study artifacts. Tools should expose the real content.

The LLM reasons. Tools observe. Evidence justifies.

Example tools:

- `parse_sap_requirements()`
- `inspect_adam_metadata()`
- `query_adam_population()`
- `extract_tlf_metadata()`
- `compare_expected_actual()`

A good finding should state what was checked, what was expected, what was observed, where the evidence came from, and who should review it. A finding without traceable evidence is not a finding; it is an unsupported opinion.

No finding without evidence.

1.7 Principle 5: Human judgment

Trust requires clear accountability.

AI-first does not mean AI-only.

The agent may identify discrepancies, assemble evidence, classify likely severity, and recommend who should review the issue. The accountable human role decides whether the finding is valid, whether the artifact should change, and how the decision should be documented.

The agent should escalate when SAP language is ambiguous, an artifact is missing, evidence conflicts, a decision requires statistical judgment, or the workflow rule does not cover the case.

Agents prepare evidence; humans make accountable decisions.

1.8 Applying the principles to SAP → ADaM → TLF review

The running example is an AI agent that reviews whether selected CSR TLF outputs derived from ADaM datasets follow the SAP.

1.8.1 Workflow objective

Verify consistency across SAP requirements, ADaM metadata and data, and TLF outputs before database lock.

1.8.2 Agent role

The agent is an independent reviewer assistant for clinical statisticians, statistical programmers, programming leads, and study reviewers.

1.8.3 Inputs

- SAP version;
- ADaM metadata snapshot;
- ADaM dataset snapshot;
- TLF package;
- workflow rules;
- prior findings and decisions, if available.

1.8.4 Example checks

Check	Evidence
SAP population exists in ADaM	population flag in ADSL or relevant dataset
SAP population definition is applied consistently	SAP population definition, ADaM flags, and TLF population counts
SAP endpoint variable exists	variable name, label, type, and codelist in ADaM metadata
Treatment group labels are consistent	SAP or TLF display labels and ADaM treatment variables
TLF title matches intended display	title and subtitle from TLF output and SAP reference
TLF N values are explainable	counts from ADaM population rules
Footnotes match analysis method	footnote text and SAP method section
Missing or imputed values are handled as specified	SAP method section, ADaM derivation metadata, and TLF footnotes
Rerun scope is justified	event and rule that triggered the run

1.8.5 Outputs

- evidence matrix;
- findings grouped by severity;
- escalation list by reviewer role;
- delta report showing fixed, new, repeated, and acknowledged findings.

1.8.6 When the workflow should stop

The agent should not continue as if the review is complete when a required artifact is missing, SAP language is ambiguous, metadata cannot be inspected, or evidence conflicts across sources.

In those cases, the correct output is an escalation or limited finding, not a confident conclusion.

1.9 Summary

The root principle is **trustworthy assistance**: the agent's work must be inspectable, repeatable, evidence-backed, and accountable.

The five derived principles are:

1. **Inspectable artifacts:** make work readable, structured, versioned, citable, and diffable.
2. **Explicit state:** remember what changed, what was checked, what was decided, and what remains unresolved.
3. **Events and rules:** trigger the agent when work changes and define what action is needed.
4. **Tools and evidence:** inspect reality with tools and require evidence for findings.
5. **Human judgment:** agents prepare the review; humans make accountable decisions.

Together, these principles keep AI-first clinical study workflows simple, reviewable, and trustworthy.

2 Onboarding an AI agent



Tip

Adding AI agent to your workflow is an onboarding problem before it is a technical one.

2.1 The first task

Before working with an AI agent, let's start with a situation you should know well.

A new member is joining your team. They finished school recently and may have taken a class on clinical trials, so they understand what a statistical analysis plan is. However, they have never seen how your organization actually works: SOPs, folder structures, naming conventions, and the dozen unwritten habits your team follows.

It is their first day. They are eager, and they want to be useful. As their mentor, you give them a first task.

Summarize the age distribution in an ongoing clinical trial's ADSL dataset.

You already know the answer to this question, or could get it in five minutes. That is the point. The task is not really about the summary statistics. It is about whether the new member can find the artifacts, follow the process, notice what looks wrong, and ask before assuming.

You are checking two things:

1. that they can reach the artifacts they need: protocol, SAP, datasets; and
2. that they work the way your team works, following established procedures, the approved computing environment, and the local conventions.

This chapter onboards exactly one new member. The only unusual thing about them is that they are an AI agent.

When a team first considers embed AI agent in their workflow, the discussion usually turns to technology: which model to choose, which tools to connect, how to install it. Those questions are real, but they are not the ones that decide whether the work can be delegated to an AI agent. The decisive questions are the same ones you answer for any new colleague. What may

they access? Which process must they follow? What do you expect that nobody wrote down? Who reviews the result before it is used?

Every one of those is an onboarding question, and every one of them is yours to answer. The model does not decide what it may read, and it does not decide who signs off on its output. You do. The rest of this chapter walks through those decisions in the order you would meet them on a real first day.

2.2 What you hand over

A new member does not arrive knowing anything about your study. You give them a starting package:

- a working environment with the software they need;
- access to the study artifacts;
- one task.

For this exercise the package is two documents, and you hand them over as two links:

Artifact	What it is
KEYNOTE-189 protocol	A Phase 3 randomized controlled trial protocol for lung cancer
kn189-synthetic-adsl.csv	A synthetic ADSL dataset

The dataset is deliberately small and deliberately imperfect. It carries the kind of untidiness a real study database has while enrolment is still open:

- **Randomization pending.** Subjects are enrolled but not yet randomized.
- **Missing data.** AGE was never recorded for some subjects.
- **Protocol deviation.** A subject's age falls outside the inclusion criteria.

You then go back to your own busy day and let the new member work independently. At the end of the day, you sit down together and review what happened.

2.3 Unstated expectations

Notice what you did not do. You did not write down which subjects to include, where the data may go, or when to stop and ask. There was no time. And if you had written it all down, you would be handing over a script rather than training someone to think.

You expect all of it anyway:

- **Read the protocol** before deciding who counts, rather than summarizing every row in the file.
- **Read the SOP** be compliant, e.g. data and code do not move somewhere they do not belong.
- **Ask** when the instruction is ambiguous, instead of guessing and moving on.
- **Report what looks wrong.** You already know about the subject outside the age range and the missing values. You want to hear about them without asking.

And you expect a short write-up at the end of the day:

- the code they ran;
- the statistics they found;
- the questions they still have.

None of this was said out loud, which is exactly how expectations arrive in real work. Onboarding AI agents like a new colleague absorbs team conventions over time, by working alongside your team and by being corrected when they get something wrong.

2.4 The new member is an agent

A language model on its own cannot do this task. It generates and interprets text; it cannot open a protocol, read a file, or execute code. An **AI agent** is the system built around a model so that it can act:

- the **harness** assembles context, calls the model, invokes tools, feeds the results back, and keeps the loop moving;
- the **model gateway** connects the harness to a chosen language model;
- **session state and memory** hold the messages and working state for the task;
- **tool interfaces** expose actions such as reading a file or running R;
- the **accessible environment** is the set of files, commands, APIs, and credentials those tools can actually reach.

The editable source for Figure 2.1 is [diagrams/intro-ai-agent.excalidraw](#).

The last item deserves the most attention. The accessible environment is the agent's equivalent of the access you granted the new member on their first day, and it is decided by you, not by the model.

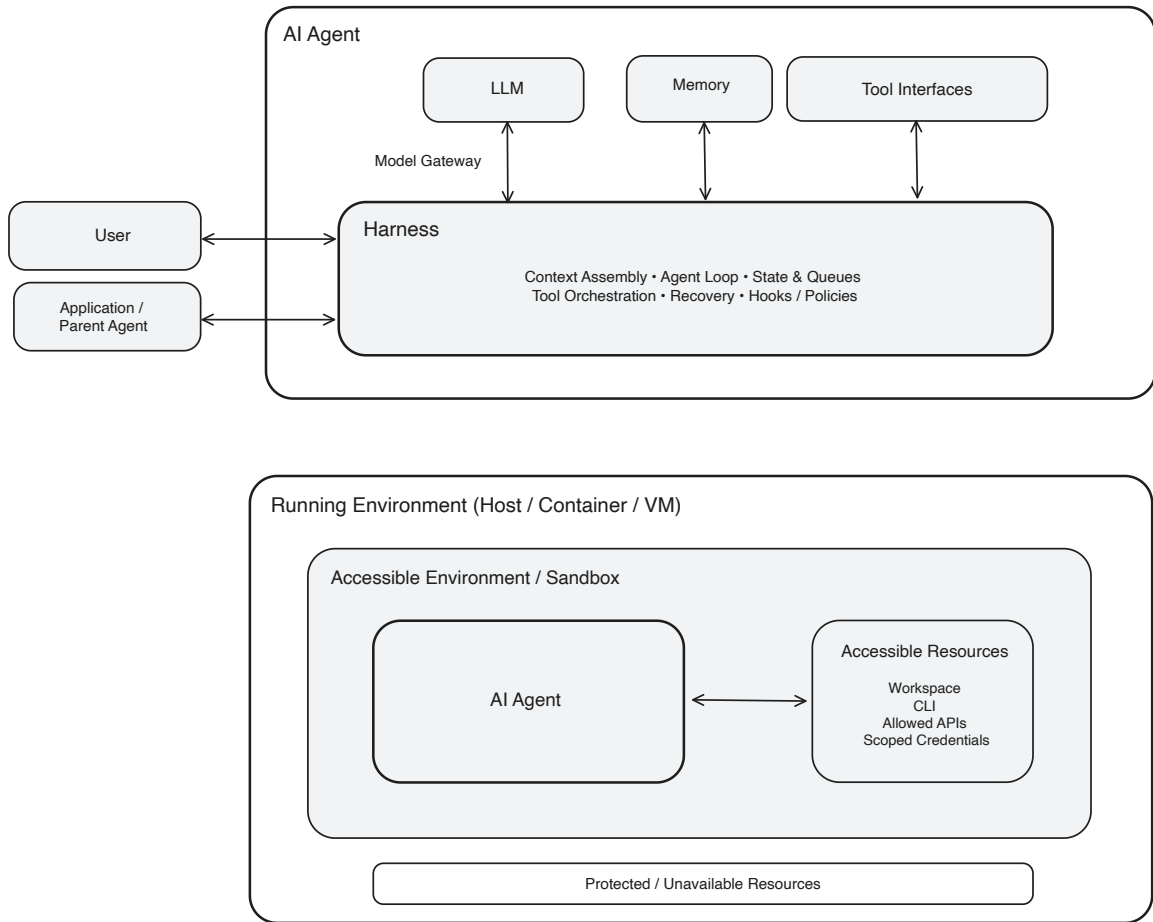
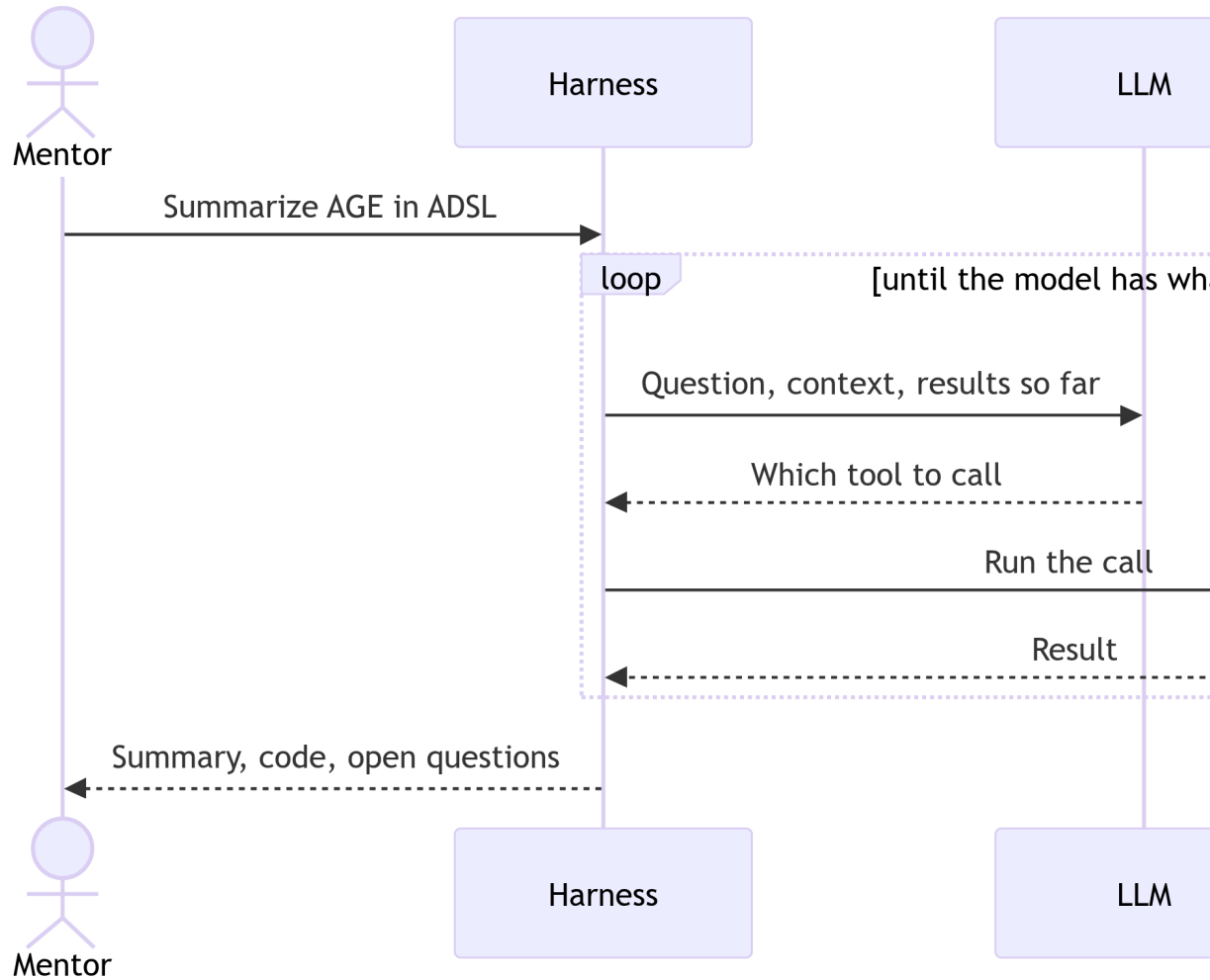


Figure 2.1: A general AI-agent architecture. The harness coordinates the language model, memory, and tool interfaces while users or applications submit requests.

2.5 Inside one turn

You ask one question. The harness turns it into a short loop and repeats that loop until the model has what it needs.



Here the loop runs twice. First to open the protocol and find the population definition. Then to run code over the subjects that definition selects.

The model never opened a file and never computed a number. It chose the population, chose the tool, and explained the answer. Everything else was done by tools, inside the environment you granted.

The model decides and explains. Deterministic software calculates.

2.6 Run the exercise

Open arena.ai/agent. It runs an agent in the browser and shows each step along the way, which makes the loop above easy to watch. Any agent that can read a document and run code will do. This one is convenient because there is nothing to install.

Give it the task the way you would give it to a person, plainly and without the answer:

```
Based on the KN189 study protocol, summarize the age distribution in the data.
```

```
protocol: https://cdn.clinicaltrials.gov/large-docs/80/NCT02578680/Prot_SAP_001.pdf
```

```
data: https://raw.githubusercontent.com/elong0527/ai4csr/refs/heads/main/exercise/day1/kn189
```

```
Important: disable all web search capability, address the question only use  
provided protocol and data.
```

! Why the last line matters

KEYNOTE-189 is a published trial. Without that instruction, the agent can search the web and answer from journal articles instead of from the documents you gave it. The numbers it returns would then describe the real study, not your dataset, and they would look entirely credible.

This is the difference between a colleague who reads your protocol and one who remembers a similar study. Only one of them is reviewing your trial.

Note what the prompt does not say: which subjects to include, and which statistics to report. The dataset carries ENRLFL and ITTFL, and the protocol defines the Intention-to-Treat population as all randomized subjects. Choosing a population is the first judgment the new member has to make, and nobody told them to make it.

Watch the interaction, not only the final answer:

- Did the agent open the protocol, or only the data?
- Which population did it summarize, and did it say which one?
- Did it write code, and did that code produce the numbers?
- Did it flag the subject aged 17, or quietly include them?
- Did it search the web anyway?

2.7 End-of-day review

This is the 1:1 you scheduled. You are not checking whether the numbers are pretty; you are checking whether the work can be trusted.

2.7.1 The numbers

You know what the answer should be, because you know the dataset. The prompt did not say which statistics to report, so compare whichever ones the agent chose to give you:

Statistic	ITT (ITTFL == "Y")	All enrolled (ENRLFL == "Y")
Nonmissing N	24	26
Missing N	1	2
Mean	50.58	51.46
Standard deviation	14.97	14.73
Q1	41.25	42.50
Median	51.00	52.50
Q3	61.25	61.75
Minimum	17	17
Maximum	76	76

Two columns, because the question has two defensible readings and only one of them follows the protocol. An answer that reports a single set of numbers without naming its population has skipped a decision rather than made one.

There is a third reading the table deliberately omits: dropping the underage subject. That one is wrong. The protocol defines ITT as *all randomized subjects*, so subject KN189-103-025 belongs in the analysis **and** belongs in an escalation. Removing them quietly is the specific failure this exercise is built to expose, because the resulting table looks entirely reasonable.

Q1 and Q3 depend on the quantile method. The values above use the default in R. The default in SAS gives 39.00 and 61.00 for the ITT population, and both are correct. Any value in that range is fine, provided the method is stated.

If the agent reports different values, that is something to investigate, not something to explain away.

2.7.2 The process

A matching number is not the same as acceptable work. Read back through the session and ask what the agent actually did: which files it touched, which commands it ran, what it sent to the model, and whether anything left the environment that should not have. A harness that keeps this record is what makes the question answerable at all.

2.7.3 The gaps

The last part of the review is the honest one. A demo that produces the right mean is not yet a validated process. Compare what you observed against your own requirements for access control, audit trail, environment qualification, and review, and note the gaps. Those gaps are the subject of the rest of the book.

2.8 Keeping data separate

The exercise also introduces a design choice that outlives it. The model rarely needs every subject-level record. Statistical software can process the raw data inside a controlled environment and return only aggregates or metadata:

```
Raw ADSL >> deterministic R/SAS/Python calculation >> aggregate result >> LLM
```

This reduces the chance that sensitive records end up in a model request. A sandbox can restrict the files, commands, network destinations, and credentials the agent can reach, and tool policies can inspect, approve, redact, or block operations at that boundary.

These mechanisms reduce risk; they do not create a guarantee. Model requests, tool output, logs, session history, temporary files, and telemetry each need deliberate treatment. An agent normally inherits whatever permissions the process that started it already holds, so isolation has to be arranged on purpose. It is never the default.

What this exercise does, and why you should not copy it

The exercise hands a browser based service a link to subject level records. That is acceptable here for one reason only: the dataset is synthetic and already public. Do not do this with study data. A hosted agent sends whatever you give it to a third party, which is the exact situation the separation above is meant to prevent. This chapter trades that protection away to keep the setup to a single browser tab. Later chapters describe the arrangement you would use instead.

The implementation belongs in later chapters. The principle belongs here:

Give the agent access to what the task requires, not everything the running environment happens to expose.

2.9 Human judgment

The agent can assemble the calculation and the explanation. It cannot be accountable for the analytical meaning. Even for a task this small, a responsible human still confirms that **AGE** and its units are appropriate, that the analysis population is the intended one, that missing values were handled sensibly, that the executed code does what it claims, and that descriptive statistics answer the question that was actually asked.

This example is deliberately trivial. The same division of responsibility matters far more once population flags, treatment groups, imputation, derived variables, or inferential methods are involved.

2.10 Takeaways

1. An agent is more than a language model: it adds a harness, state, tools, and an accessible environment.
2. Onboarding an agent raises the same questions as onboarding a person: access, process, and unstated expectations. Answering them is your job, not the model's.
3. The model decides and interprets; deterministic statistical software calculates.
4. The harness makes the loop observable, which is what lets you review the work rather than just the answer.
5. A plausible-looking result is not a verified one. Human review remains accountable for the statistical meaning and the final use.

Later chapters develop context construction, tool design, memory, security, governance, reproducibility, and validation in more depth.

Part II

Documentation as code

3 Documentation overview

 Objective

TBD

3.1 TBD

TBD

Part III

Analysis and reporting

4 Reporting overview

 Objective

TBD

4.1 TBD

TBD

References